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Departure Times in Y-Shaped Traffic Networks with Multiple Bottlenecks

By TERRY E. DANIEL, EYRAN J. GISCHES, AND AMNON RAPOPORT*

The economic analysis of congested traffic networks has a venerable history. Gaining prominence in the classic debate over social costs between A. C. Pigou and Frank H. Knight (1924), and later on in the seminal article in this journal by William Vickrey (1969), the problem continues to attract academic and public interest.¹ The research reported in our paper follows directly from Vickrey's model of the decisions commuters make as to when to depart from a common origin and travel to a common destination along a single road. In Vickrey's model, there is a single bottleneck on the road with a fixed and known capacity, and a common desired arrival time at the common destination. If the arrival rate at the bottleneck exceeds the bottleneck capacity, as it is likely to occur if most commuters depart at a time that would otherwise get them to the destination at the desired arrival time, then a queue forms. Commuters then face a trade-off. A commuter may choose to depart early and pay the penalty of arriving at her destination too early (with respect to an exogenously determined desired arrival time), depart late and pay the (much heavier) penalty of arriving too late, or depart at the peak hour when longer queues form and delays are high.² In Vickrey's model, each commuter independently chooses her departure time to minimize her travel costs, which are assumed to be linear functions of travel time and the time that she is either early or late in reaching the destination (these latter times are hereafter referred to as *schedule delays*). Departure times are in equilibrium when no commuter has an incentive to unilaterally alter her departure time decision, given those of the other commuters.

Paradoxes in Traffic Networks.—Alternative formulations and extensions of Vickrey's model have been developed by, among others, Chris T. Hendrickson and George Kocur (1981); Mike J. Smith (1983); Carlos F. Daganzo (1985); Masao Kuwahara (1990); Richard Arnott, André de Palma, and C. Robin Lindsey (1990, 1993a, b, 1999); and Hironori Otsubo and Rapoport (2009). Of particular relevance to our paper is the model by Arnott, De Palma, and Lindsey (1993a, hereafter ADL), who considered a Y-shaped traffic network comprising *two* upstream bottlenecks with service capacities, s_1 and s_2 , and a single downstream bottleneck with service capacity, s_d . In their model, two groups of commuters travel along the Y-shaped corridor from home to work,

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¹For example, a *New York Times* article (Daniel Gross, "What's the Toll? It Depends on the Time of Day," February 11, 2007) discussing federal budget initiatives to stimulate the management of traffic delays on American roadways heralded the near future as the "golden age of congestion pricing."

²Examples abound. In deciding when to leave for the airport for a designated flight, one may weigh the expected delays at checkpoints in counters and in security lines against the costs of arriving too early or too late for the flight.

one entering each arm and passing through the corresponding upstream bottleneck, as well as the downstream bottleneck that is common to both groups. Imposing linear travel and schedule delay costs, ADL then reported a remarkable and counterintuitive result, namely, that for certain combinations of bottleneck capacities, per unit schedule delay costs, and per unit travel time costs, *expanding* one of the upstream bottlenecks may induce the commuters to alter their departure times in such a way as to *increase* the sum of total traveling and schedule delay costs.³ Our study reports the results of an experiment designed to study the implications of this theoretical result in the controlled environment of the laboratory.

The result reported by ADL seems paradoxical, as the usual remedy for combating congestion is to expand the road system (increase service capacity). Transportation researchers have identified additional paradoxes in which this remedy fails (Arnott and Kenneth Small 1994). The very well known *Pigou-Knight-Downs* (PKD) paradox concerns a simple network with a common origin and a common destination connected by two roads, a direct road containing a narrow bridge, and a wider, more circuitous, road. For certain parameter combinations (number of commuters, bridge capacity), expanding the bridge's capacity has no effect on travel time. A second, more perverse, paradox is the *Downs-Thompson* paradox. It resembles the previous paradox except that the alternative to taking the congested road with the narrow bridge is replaced by a privately operated train line. Both paradoxes are illustrated numerically by Arnott and Small (1994). The third paradox, arguably the most well known to engineers and computer scientists, is due to Dietrich Braess (1968). It consists of showing that, in equilibrium, adding one or more links to a traffic network, thereby allowing for more routes between a common origin and a common destination, *may* cause total travel time to increase. In all of these paradoxes, the commuters depart from the origin of the network simultaneously. In the *Bottleneck Paradox* of ADL, which is the subject of our paper, the number of network users, origins, common destination, travel routes, and cost structures are all *fixed*, but departure times are endogenously determined.

Previous Empirical and Experimental Research.—All of these paradoxes occur in abstract networks with alternative routes or modes of transportation, on at least one of which unpriced congestion occurs. The equilibrium solution (called *user equilibrium* in the context of traffic networks) is inefficient because network users make their decisions selfishly without taking into account the negative externalities they impose on others. After reviewing and numerically illustrating some of these transportation paradoxes, Arnott and Small (1994, 451) raised the question: “Are these paradoxes more than intellectual curiosities?” Empirical data that either reject or support the paradox are very hard to come by, as the assumptions underlying these models cannot exactly be met. As early as 1970, John D. Murchland noted with regard to the Braess paradox: “It seems to me that the importance of the Braess’ paradox for practical networks will only become apparent when sufficiently accurate congested traffic assignment calculations become available and the phenomenon emerges, or fails to emerge, during systematic searches for optimal link additions” (1970, 393). Few of these systematic searches have been forthcoming. Perhaps the most frequently cited example is the one by Walter Knödel (1969), who remarked that major road improvement in the center of Stuttgart failed to yield the benefits expected. They were only obtained when a cross street was subsequently withdrawn from traffic use. Intriguing as it may be, this is only anecdotal evidence that does not meet the requirement for “systematic searches for optimal link additions.”

An alternative approach that has emerged in economics is to test the implications of the equilibrium analyses of various traffic networks and compare user behavior with the equilibrium

³ This result could be used as the theoretical justification for the practice of regulating the flow of traffic at freeway on-ramps (“metering”).

solution *experimentally*. This approach consists of simulating simple traffic networks in the controlled environment of the laboratory, having users choose routes in these simulated traffic networks before and after they are subjected to changes in the cost function or structure of the network, and searching for the emergence of systematic and replicable patterns of behavior. If they emerge, then it is of significant interest to determine whether these behavioral patterns support equilibrium play and the paradoxical results it may imply. Experimental studies that subscribe to this approach have recently been reported by Erel Avineri (2006); Yasunori Iida, Takamasa Akiyama, and Takashi Uchida (1992); Laurent Denant-Boèmont and Romain Petiot (2003); Dirk Helbing (2004); Helbing, Martin Schönhof, and Daniel Kern (2002); Yannick Gabuthy, Matthieu Neveu, and Denant-Boèmont (2006); Reinhard Selten et al. (2004, 2007); Kerstin Schneider and Joachim Weimann (2004); Avineri and Joseph N. Prashker (2005); Rapoport et al. (2008); John Morgan, Henrik Orzen, and Martin Sefton (2009); Rapoport, Vincent Mak, and Rami Zwick (2006); and Anthony Ziegelmeyer et al. (2008).

With the exception of the experiments by Gabuthy et al. (2006), Schneider and Weimann (2004), and Ziegelmeyer et al. (2008), all of these experiments have focused on *route choice*, not on *departure time* in networks that are not susceptible to structural changes. Departure times are considered in Gabuthy et al., but model parameters are chosen so that queues do not form in equilibrium. Schneider and Weimann tested a road-pricing model formulated by Arnott, De Palma, and Lindsey (1990). In their model, a fixed number of commuters have to travel from a common origin, O , to a common destination, D , on a single road and pass through a single bottleneck with a fixed capacity s . The Bottleneck Paradox of ADL does not arise in such a simple network. Ziegelmeyer et al. (2008) reported two laboratory experiments designed to study the impact of public information about past departure rates on travel costs and congestion levels. Their focus is on population size, relative cost of delay, and outcome information about previous congestion rates, not on the paradoxical results of ADL. Hani Mahmassani, Gang-len Chang, and Robert Herman (1986) and Mahmassani and Chang (1987) conducted a series of laboratory studies of departure time choice behavior in a simple congested network. They compared their results to a model in which commuters were assumed to be boundedly rational to the degree that they were “satisfied” with arrival times in a band around their desired arrival time. Their model and experimental methods differ substantially, however, from those of our study.

The rest of the paper is organized as follows. Section I presents our model of the Y-shaped network with bottlenecks. It is similar to the ADL model with the following major exception: we allow for late arrivals (at high cost) to the common destination with respect to a common desired arrival time (as does Masao Kuwahara (1990)). Section II begins with an exposition of the Nash equilibrium solution for this extension of the ADL model. It then introduces the corresponding finite-agent model that we later test in the experiment. Modifications of the equilibrium solution necessary for the finite-agent case are discussed. Section III presents the experimental method and results. Our contribution is threefold. First, and most important, we show experimentally that the Bottleneck Paradox is not just a theoretical curiosity. Using a between-subject design, we demonstrate that—for the cost structure in the present study—an increase in the capacity of the upstream bottleneck causes a significant shift in the departure time distribution and, consequently, a significant increase in commuters’ total travel cost. Second, we find that observed departure times on the aggregate level are in close agreement with the equilibrium solution. Third, we show that experience in iteratively traversing the network matters. As the network users who have to pass through two bottlenecks gain more experience with the network, their mean payoff increases and the between-user variability in payoff decreases sharply. Finally, we show that this systematic change in aggregate behavior over time can be accounted for by a simple reinforcement-learning model. Section IV concludes with a discussion regarding the significance and generality of our results.

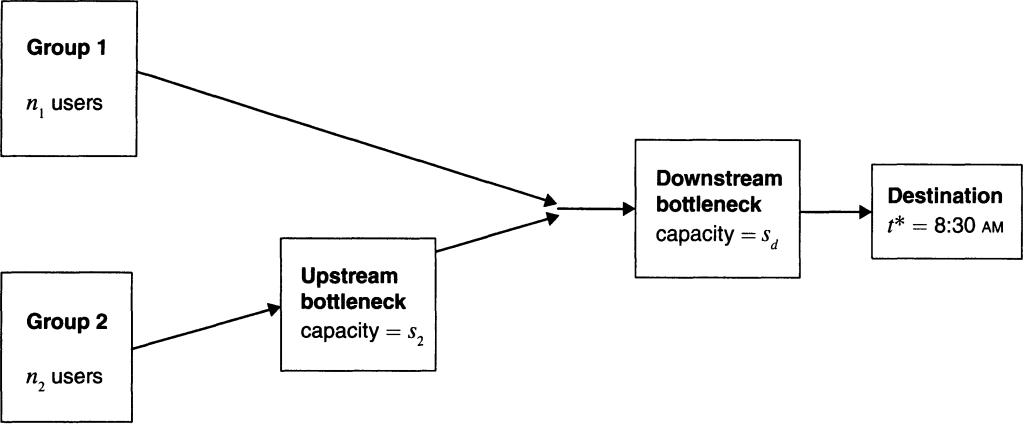


FIGURE 1. TRAFFIC NETWORK WITH TWO GROUPS AND TWO BOTTLENECKS

Notes: Group 1 passes through the downstream bottleneck with capacity s_d , and group 2 passes through the upstream and downstream bottlenecks with respective capacities s_2 and s_d . Both groups wish to arrive at the common destination at 8:30 AM. Departure times are endogenously determined.

I. The Model

We consider the Y-shaped traffic network that is illustrated in Figure 1. There are two groups of network users who commute daily from their neighborhoods of origin to a common destination. On their journey they encounter bottlenecks that have the potential to introduce delay into their travel. The n_1 users in group 1 pass through only a downstream bottleneck common to both groups whereas the n_2 users in group 2 first must pass through an upstream bottleneck and then join the users from group 1 in traversing the downstream bottleneck.

The upstream and downstream bottlenecks have nonstochastic capacities s_2 and s_d , respectively, where capacity is measured by the maximum number of network users who can pass through per unit time. If the arrival rate of commuters at either of the two bottlenecks exceeds its capacity, a queue forms behind this bottleneck. This is the only source of congestion on the network.⁴ The decision variables in the model are the individual departure times, and these are independently and endogenously determined.

There are two sources of cost that are common to the network users of groups 1 and 2. There is the cost per unit of *travel time*, which we denote by α . Without loss of generality, we assume that travel time is zero except for the delay experienced in the bottlenecks. There are also the costs of arriving either too early or too late with respect to a common, exogenously determined, desired arrival time, t^* . The parameter β denotes the cost per unit time of *early arrival*, and the parameter γ the cost per unit time of *late arrival*. Like ADL, we assume $\gamma > \alpha > \beta$. This inequality generally holds in practice. Thus, each commuter is presented with a departure time choice problem that involves trading off delays in bottlenecks, which are clearly a function of the departure times of the other commuters, with desired arrival time at the common destination.

⁴ The two bottlenecks are assumed to be sufficiently separated so that the physical length of a queue behind the downstream bottleneck does not interfere with the upstream bottleneck.

Let $T_g(t)$ be the total travel delay at bottlenecks for a member of group g , who departs home at time t . Then, the corresponding travel cost of such a commuter from group g , who departs home at time t , can be written as

$$(1) \quad C_g(t) = \alpha T_g(t) + \beta \max[0, t^* - (t + T_g(t))] + \gamma \max[0, (t + T_g(t)) - t^*].$$

Section II presents equilibrium results for a case in which the Bottleneck Paradox emerges in this model. As the capacity of the upstream bottleneck is increased, commuter travel costs also increase.

II. The Equilibrium Solution

A. The Infinite-Agent Model

In this section, we present the Nash equilibrium distribution of departure times for the model of the Y-shaped network presented in Section I. This equilibrium is similar to that in ADL with one significant change; namely, the cost of late arrivals, γ , is no longer infinitely large. Consequently, it is optimal in our model for some commuters to arrive late at their common destination. The ADL model is a “flow” model in that the commuters in each group are assumed to constitute an infinite set of infinitesimally small elements with n_1 and n_2 being measures of the size of the two groups. As a result, if a bottleneck has a capacity of commuters s per hour, then n_1/s hours would be required for all of the members of group 1 to pass through that bottleneck, even though each commuter would require only an infinitesimally small amount of time to do so. This model is a useful approximation for real traffic networks with large numbers of commuters. Later in this section we discuss the adjustments necessary to the equilibrium results for a model with a finite and relatively small number of agents (as required for laboratory testing with human participants) who take a finite amount of time to traverse a bottleneck.

Let $f_g(t)$ represent the distribution of departure times for each group g ($g = 1, 2$) of users. If τ_g is the set of times for which $f_g(t)$ is greater than 0 (the set of times over which users in group g depart), then $\int_{\tau_g} f_g(t) dt = n_g$. The distributions of departure times of the two groups, $f_1(t)$ and $f_2(t)$, constitute a Nash Equilibrium if and only if the costs of departing at any time in the sets τ_1 and τ_2 are constant for each group and less than the costs of departing at all other times,

$$(2) \quad C_g(t) \begin{cases} = \bar{C}_g & \text{for } t \in \tau_g \\ > \bar{C}_g & \text{for } t \notin \tau_g \end{cases} \quad g = 1, 2.$$

For the Y-shaped network, because group 1 users face no upstream bottleneck, $\bar{C}_1 \leq \bar{C}_2$. Following ADL, we illustrate the Bottleneck Paradox for the case where $\bar{C}_1 < \bar{C}_2$ and $s_2 \geq s_d$.

Graphs of the equilibrium cumulative distributions of departure times of the two groups of (infinitely many) network users are exhibited in Figures 2A and 2B for the two sets of model parameters⁵ that will be used in the experiments. Figure 2A depicts results for the situation in which the upstream bottleneck has a relatively low capacity, and Figure 2B depicts the same

⁵ For comparability, we have chosen parameters that are proportional to the ones used in the numerical examples in ADL. The only changes are to multiply population measures and bottleneck capacities by 16 so as to correspond to the discrete quantities used in the experiments and to introduce the finite cost of being late. The relative sizes of the cost parameters are values commonly used in the transportation modeling literature.

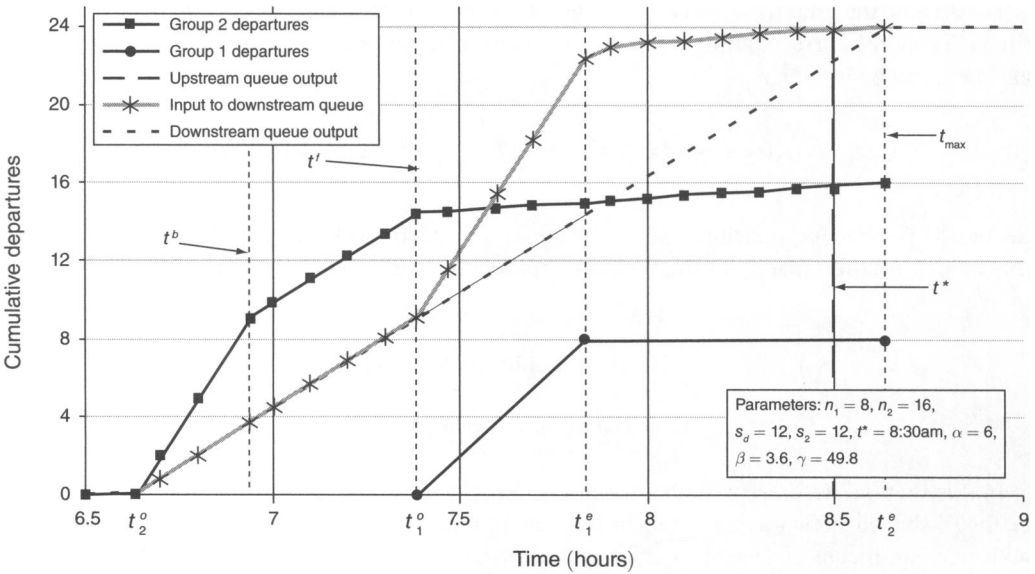


FIGURE 2A. EQUILIBRIUM DISTRIBUTION OF CUMULATIVE DEPARTURE TIMES FOR LOW UPSTREAM CAPACITY FOR ADL FLOW MODEL WITH LATE ARRIVALS PERMITTED (FINITE LATE COST)

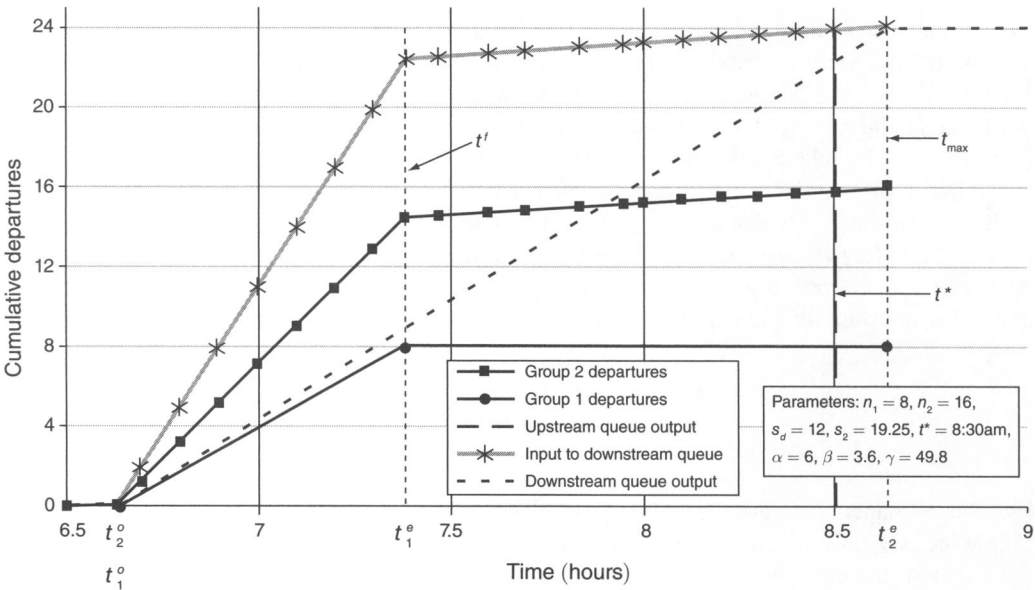


FIGURE 2B. EQUILIBRIUM DISTRIBUTION OF CUMULATIVE DEPARTURE TIMES FOR HIGH UPSTREAM CAPACITY FOR ADL FLOW MODEL WITH LATE ARRIVALS PERMITTED (FINITE LATE COST)

network with a higher upstream bottleneck capacity. Derivation of the results shown in these graphs is contained in a working paper available from the authors. In the present paper we confine ourselves to the insights, several of them due to ADL, that drive the derivations.

We begin with an analysis of the low capacity situation as illustrated in Figure 2A. We will later observe that the high capacity case illustrated in Figure 2B is simply an extreme case and the same analysis will apply. The first thing to observe in Figure 2A is that, in equilibrium, some network users arrive late (after t^*) and that they are all from group 2. The reason for this will become clear below. Let t_g^o and t_g^e denote, respectively, the departure times of the first and last members of each group g . By equating (as the equilibrium solution requires) the travel cost of the first user from group 2 to depart, who encounters no delay and only an early arrival cost, with the travel cost of the last user to depart from group 2, who also faces no delay but incurs a late arrival cost, it is relatively straightforward to show that the departure time of the last user to arrive at the destination, $t_{\max}$, must satisfy the equation

$$(3) \quad t_{\max} = t_2^e = t^* + \left(\frac{\beta}{\beta + \gamma} \right) \left(\frac{n_1 + n_2}{s_d} \right),$$

and also that the departure time of the first commuter to depart, t_2^o , must be

$$(4) \quad t_2^o = t_{\max} - (n_1 + n_2)/s_d.$$

Between t_2^o and $t_{\max}$, group 2 users (Figure 2A) depart at *three* different rates resulting in the following cumulative distribution of departure times (as shown in Figure 2A):

$$(5) \quad (a) \quad F_2(t) = \frac{\alpha}{(\alpha - \beta)} s_d(t - t_2^o) \quad \text{for } t_2^o \leq t < t^b$$

$$(b) \quad F_2(t) = s_2(t - t^b) + \left(n_2 - \frac{(n_1 + n_2)\beta}{\beta + \gamma} - \frac{n_1 s_2(\alpha - \beta)}{\alpha s_d - (\alpha - \beta)s_2} \right) \quad \text{for } t^b \leq t < t^f$$

$$(c) \quad F_2(t) = \frac{\alpha}{\alpha + \gamma} s_d(t - t^f) + \frac{n_2\gamma - n_1\beta}{\beta + \gamma} \quad \text{for } t^f \leq t \leq t_{\max},$$

where the first breakpoint in this distribution, $t^b = t_2^o + [(\alpha - \beta)/\alpha s_d][(n_2 - ((n_1 + n_2)\beta/(\beta + \gamma)) - (n_1 s_2(\alpha - \beta)/(\alpha s_d - (\alpha - \beta)s_2))]$, is the time at which group 2 commuters leaving the upstream bottleneck encounter the first group 1 commuter at the downstream bottleneck, and the second breakpoint, $t^f = t^* - (\gamma/\alpha)[\beta(n_1 + n_2)/((\beta + \gamma)s_d)]$, is the departure time of the first group 2 commuter who will arrive late at the destination.

The initial rate specified in equation (5a) is greater than the upstream bottleneck capacity for the parameters used in our model, and as a result a queue forms at the upstream bottleneck. This departure rate reflects the cost trade-off between reduced early arrival times at the destination and increased time waiting in queues. The departure rate for group 2 falls to a rate that is equal to the capacity of the upstream queue, s_2 , at t^b and the upstream queue length then stays constant for the entire period that these users encounter group 1 users at the downstream bottleneck. This is clearly a necessary condition for equilibrium because the only difference between the two groups of users is the cost that group 2 users encounter in the upstream bottleneck.⁶

⁶In equilibrium, $C_1(t)$ and $C_2(t)$ are both constant for all departure times in τ_1 and τ_2 , respectively; hence $C_2(t) - C_1(t)$ must be constant and this can be achieved only if the upstream queue delay is constant for all group 2 users who arrive at the downstream bottleneck during the period when group 1 users are departing.

Between t^f and $t_{\max}$, group 2 users depart at the much lower (assuming large γ) rate shown in equation (5(c)). There are $s_d(t_{\max} - t^*)$ users in this last set of group 2 users, and they arrive late at the destination at times between t^* and $t_{\max}$. Their rate of departure is determined by the trade-off between the costs of waiting in line and of being late.

As can be seen in Figure 2A, group 1 users depart at times in the mid-range of the departure times for group 2 users. As discussed earlier, group 1 departures must take place while there is a queue at the upstream bottleneck. If not, group 1 and group 2 users departing at such a point in time would have the same cost (because of no delay cost at the upstream bottleneck) in violation of $\bar{C}_1 < \bar{C}_2$. In particular, the cumulative distribution of departure times for group 1 is given by

$$(6) \quad F_1(t) = \left(\frac{\alpha}{\alpha - \beta} s_d - s_2 \right) (t - t_1^o) \quad \text{for } t_1^o \leq t < t_1^e,$$

where the last departure from group 1, $t_1^e = t_2^o + [(n_2 - sd(t_{\max} - t^*)/s_2)]$, occurs when the upstream queue begins to empty, and the first departure in group 1 occurs at $t_1^o = t_1^e - [(n_1/((\alpha/(\alpha - \beta))s_d - s_2))]$, so as to give all group 1 commuters a constant cost over τ_1 .

It is instructive to consider the experiences of a few of these commuters. For example, with the parameter set illustrated in Figure 2A, the first commuter in group 2 departs at 6:38 AM, faces no queues or delays at either bottleneck, and arrives at the destination very early. However, because the departure rate of this first cohort is much greater than the capacity of the upstream bottleneck, a commuter leaving just 16 minutes later at 6:54 AM encounters a long queue and a delay of nearly 24 minutes at the first bottleneck. Because the capacities of both bottlenecks are the same, the upstream bottleneck regulates flow such that there is no further delay at the downstream bottleneck. Any group 2 users who depart after 6:56 AM encounter additional delays at the downstream bottleneck because by the time they get there, group 1 users will have started joining the flow and queues begin to build up. For example, a commuter in group 2 who leaves at 7:06 AM spends 26 minutes in the upstream queue and then another 16 minutes in the downstream bottleneck—for a total delay of 42 minutes. For group 2 commuters leaving at the tail end of the rush period, the delays become shorter but they are always late. A commuter who leaves at 8:10 AM, for example, faces no upstream delay but a delay of 25 minutes at the downstream queue and arrives at the destination 5 minutes late at 8:35 AM. Group 1 commuters leave over a very compact segment of time in the middle of the commute period. The first group 1 person leaves at 7:23 AM, faces no queue at the downstream bottleneck, and experiences no delay. Thereafter, the joint flow of commuters from both groups results in ever-longer queues at the downstream bottleneck. The final group 1 commuter departs at 7:50 AM and is delayed 40 minutes at the downstream bottleneck, arriving at her destination exactly on time at 8:30 AM.

The departure times and distributions in equations (3) through (6) that are illustrated in Figure 2A yield the following costs. All group 1 users have a commuting cost, in equilibrium, of

$$(7) \quad \bar{C}_1 = \alpha \left(\frac{\gamma}{\beta + \gamma} \left(\frac{n_1 + n_2}{s_d} \right) - \frac{n_2}{s_2} + \frac{\beta}{\beta + \gamma} \left(\frac{n_1 + n_2}{s_2} \right) \right),$$

and for all group 2 users the commuting cost is

$$(8) \quad \bar{C}_2 = \beta \left(\frac{\gamma}{\beta + \gamma} \left(\frac{n_1 + n_2}{s_d} \right) \right).$$

The resulting total system cost for all users in both groups is, therefore,

$$(9) \quad TC = n_1 \bar{C}_1 + n_2 \bar{C}_2.$$

From the expressions above it is easy to verify that $\partial TC / \partial s_2 > 0$ under the conditions of our model, $\bar{C}_1 < \bar{C}_2$, and $s_2 \geq s_d$. This means that increasing the capacity of the upstream bottleneck, as is illustrated in Figure 2B, results in an increase of total network costs. This is the Bottleneck Paradox discussed by ADL.

As the capacity of the upstream bottleneck increases (from $s_2 = 12$ to $s_2 = 19.25$), it becomes less of a constraint on the behavior of the group 2 commuters. Equations (5(a, b, c)) still describe the cumulative distribution of departures, but t^b approaches t_2^o and 5(a) applies over an increasingly short period. Equation (6) also still applies to the group 1 commuters but their initial departure time, t_1^o , moves closer to that of the group 2 commuters. Eventually the capacity of the upstream bottleneck ceases to be a binding constraint and the network behaves as if only the downstream bottleneck existed. Figure 2B depicts the equilibrium behavior of the flow model for a situation that is just incrementally shy of this point.⁷

B. The Finite-Agent Model

Because a laboratory experiment necessitates a finite and relatively small set of commuters, we next shift our focus from the infinite-agent flow model to a corresponding finite-agent model. The parameters for our experiments have been chosen to be proportional to those used in ADL. Specifically, we set $n_1 = 8$, $n_2 = 16$, $s_d = 12$, $t^* = 8:30$, $\alpha = 6c/hr$, $\beta = 3.60c/hr$, and $\gamma = 49.8c/hr$, with $s_2 = 12$ in the case of low upstream capacity (later called the LOW condition in our experiments) and $s_2 = 20$ representing high upstream capacity (the HIGH condition).⁸ These values result in a finite-agent model ($n = 24$) in which the set of agents in each district requires the same total time to traverse the bottlenecks as in ADL's flow model.

Figure 3A is a bar graph illustrating a distribution of departure times and the composition of each trip for 24 commuters in the discrete-agent model in the LOW condition (corresponding to the flow model results in Figure 2A). The departure time distributions shown in Figure 3 were computed using a mathematical programming optimization algorithm to find the set of departure times that minimize total commuting costs while keeping the costs of commuters within each group the same. Beginning at the left of each bar we see the departure time of the agent followed by the times the agent arrives at the upstream and downstream bottlenecks and finally at the destination. Overlaid on the bar graphs of the 16 members of group 2 are corresponding graphs for the 8 members of group 1 (who start roughly 1 hour later). As can be seen, the departure distributions in Figures 2 and 3 are very similar, with the main difference being the finite processing time required in the bottlenecks in the finite-agent case depicted in Figure 3. This adds a fixed amount of time to the total distribution of commuting times. The discrete distribution exhibited in Figure 3A is not a pure-strategy Nash equilibrium (despite giving all commuters equal costs) because any commuter could depart incrementally later (up to the time of the next departing commuter) and reduce his or her cost. This will be true of any discrete set of departure times that

⁷ Specifically, when $s_2 > \alpha s_d (n_2 \gamma - n_1 \beta) / ((\alpha - \beta)(n_1 + n_2) \gamma)$, it ceases to be a binding constraint.

⁸ In the model, the costs are expressed in c/hr , where c is a unit cost. However, in the experiment the corresponding units presented to the subjects were "points" where $1c = 100$ points. In the finite-agent model, the parameters chosen for the HIGH condition still represent a situation in which the upstream bottleneck is a binding constraint.

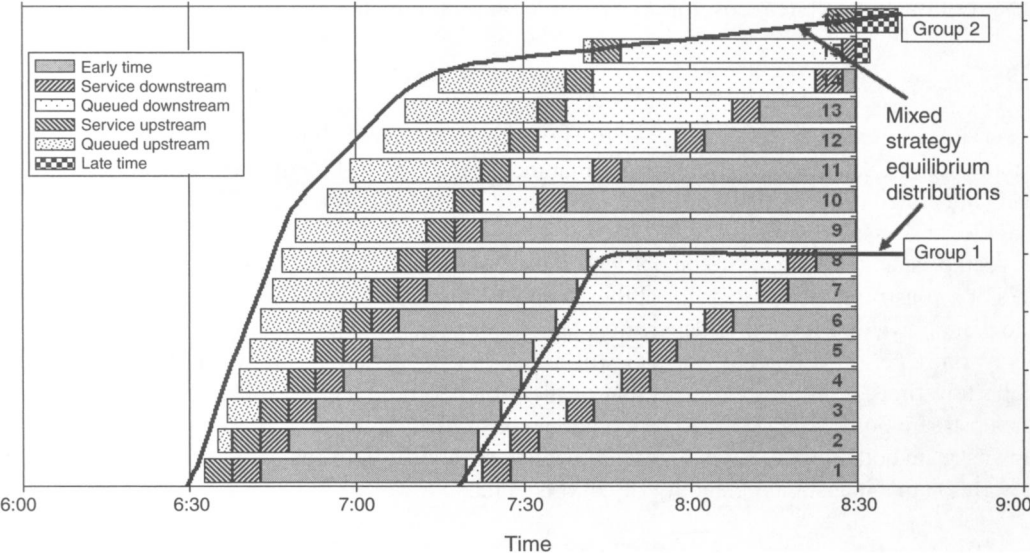


FIGURE 3A. DISCRETE AND MIXED-STRATEGY DEPARTURE AND QUEUING TIMES—LOW UPSTREAM CAPACITY

Notes: Bar graph shows composition of trip for each group 2 commuter (1–16). Starting later and overlaid are trips for group 1 commuters (1–8).

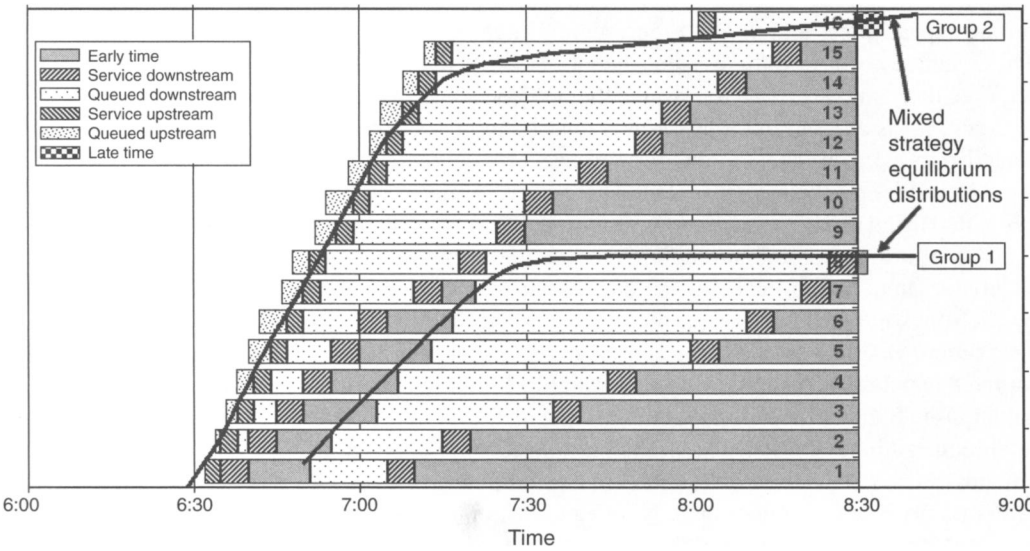


FIGURE 3B. DISCRETE AND MIXED-STRATEGY DEPARTURE AND QUEUING TIMES—HIGH UPSTREAM CAPACITY

results in equal cost for all players. Nash stability with equal cost will occur only in this finite-agent, continuous time model when the agents use mixed strategies.⁹

⁹ In general, pure-strategy equilibria exist only in discrete models in which the number of potential departure times is small relative to the population size, and even then depend on the cost parameters and the bottleneck capacities.

TABLE 1—EQUILIBRIUM DEPARTURE TIMES AND COSTS (c/hr) FOR
CONTINUOUS AND DISCRETE MODELS

	ADL flow model	Discrete programming model	Mixed-strategy simulation
<i>LOW upstream capacity</i> ($s_2 = 12, n_1 = 8, n_2 = 16$)			
Start time group 2 (t_2^g)	6.63	6.55	6.47
Start time group 1 (t_1^g)	7.39	7.33	7.31
Start late arrivals (t^f)	7.38	7.26	7.17
Last departure group 1 (t_1^e)	7.83	7.70	7.75
End time group 2 ($t_{\max}$)	8.63	8.63	8.64 ^a
Group 1 cost/person ($\bar{C}_1$)	4.00	4.52	4.49
Group 2 cost/person ($\bar{C}_2$)	6.71	7.42	7.64
Total system cost	139.36	154.88	158.16
<i>HIGH upstream capacity</i> ($s_2 = 20, n_1 = 8, n_2 = 16$)			
Start time group 2 (t_2^g)	6.63	6.53	6.48
Start time group 1 (t_1^g)	6.63	6.85	6.69
Start late arrivals (t^f)	7.38	7.25	7.17
Last departure group 1 (t_1^e)	7.38 ^b	7.38	7.49
End time group 2 ($t_{\max}$)	8.63	8.58	8.64 ^a
Group 1 cost/person ($\bar{C}_1$)	6.71	6.71	6.85
Group 2 cost/person ($\bar{C}_2$)	6.71	7.41	7.55
Total system cost	161.04	172.24	175.60

^a Last arrival is random in this model. The table reports the expected value.
^b At HIGH capacity, a small fraction of group 1 commuters depart after this time and are late.

The solid curves in Figure 3A display departure time distributions obtained using simulation of symmetric mixed strategies to find least-cost departure distributions which are Nash stable (result in equal expected costs for all commuters departing at times in the distribution and higher costs for all times outside the distribution).¹⁰ The similarities of the two distributions in Figure 3A (and in Figure 3B for the case of HIGH upstream capacity) make it relatively immaterial for the Bottleneck Paradox as to which is chosen as the theoretical benchmark for comparison to the experimental results. We will use the symmetric mixed-strategy model but note later in the paper that actual individual play suggests that many players appear to use strategies that are similar to the (nonequilibrium) discrete model.

The upper panel of Table 1 compares the departure times for the ADL model described earlier in this section and illustrated in Figure 2A, and both of the mixed-strategy and finite-agent models shown in Figure 3A. Total commuting costs per agent are also shown for each of the three models. The lower panel of Table 1 provides the same information for the case of higher upstream capacity ($s_2 = 20$).

Comparison of the two panels in Table 1 shows that each of the three theoretical models yields a strong paradoxical effect. As the capacity of the upstream bottleneck is increased, total commuting costs also increase. (Compare the group 1 cost/person: $\bar{C}_1 = 4.00$ for the LOW condition

Ziegelmeyer et al. (2008) contains an instructive analysis of such situations in a single bottleneck network including a nice two-person, two-time-slot example.
¹⁰ A detailed appendix to this paper is available on the AER Web site (<http://www.aeaweb.org/articles.php?doi=10.1257/aer.99.5.2149>). It describes the mathematical programming and numerical models used to find the discrete and symmetric mixed-strategy solutions. The appendix also illustrates the convergence of the predictions of both models to those of the ADL flow model as the number of commuters increases.

versus 6.71 for the HIGH condition for the ADL model, 4.52 versus 6.71 for the discrete programming model, and 4.49 versus 6.85 for the symmetric mixed-strategy simulation). The experiment outlined in the next section is designed to test whether this paradox is realized in the laboratory.

III. The Experiment

A. Method

Subjects.—Two hundred and forty subjects, in approximately equal proportions of men and women, participated in the experiment. The subjects were primarily business administration undergraduate students who volunteered to participate in a group decision-making experiment with payoff contingent on performance. They were recruited using class announcements and electronic bulletin board postings. The participants were divided into 10 equal-size groups of 24 members each (5 groups in each of 2 conditions). Each group participated in a single session consisting of 55 trials (trials 1–5 were for practice only) that lasted about 95 minutes, including 25 minutes of instruction time and practice.

Procedure.—The experiment used a between-subject design with two experimental conditions and five groups in each condition (total of ten sessions). We refer to these two treatments as conditions HIGH ($s_2 = 20/\text{hr}$ resulting in 3 minutes per user) and LOW ($s_2 = 12/\text{hr}$, resulting in 5 minutes per user) in accordance with the capacity of the upstream bottleneck. The experiment was conducted at the Economic Science Laboratory (ESL) at the University of Arizona. The ESL includes multiple networked PC terminals separated by partitions. Upon arrival to the lab, each subject chose a poker chip from a bag containing 24 chips to determine his or her seating. Any form of communication between subjects was strictly forbidden. Once randomly seated in their cubicles, the subjects proceeded to read the instructions (see Appendix A for the instructions for condition LOW) at their own pace. Questions asked during this period were privately answered by the experimenter.

The parameters for the two experimental conditions were chosen to correspond to the two cases (LOW and HIGH) for which model results were presented in the previous section. Two transformations of those values were necessary. First, the monetary values for the travel costs were expressed in terms of “points” when presented to the subjects (and then converted into money at the end of the experiment at the rate $\$1 = 750$ points). Second, on each trial the subjects were given a reward of 1,000 points for reaching their destination so as to generally (but not always) allow them to make profitable departure time decisions. Thus, the parameters in condition HIGH assumed the values $n_1 = 8$, $n_2 = 16$, $s_d = 12/\text{hr}$ (5 min per user), $s_2 = 20/\text{hr}$ (3 min per user), $\alpha = 10$ pts/min, $\beta = 6$ pts/min, $\gamma = 83$ pts/min, and $t^* = 8:30$ AM. Condition LOW used the same parameter values with the only exception that $s_2 = 12/\text{hr}$ (5 min per user). In order to approximate continuous time, departure times for the participants were restricted to the nearest full second over the time interval 4:00 AM to 8:30 AM—a total of 16,200 possible decisions (see the decision screen in the instructions in Appendix A).

Subjects were promised \$5 for their participation in the experiment. At the end of the session, the cumulative earnings across all 50 trials were added to the initial participation fee to determine the payoff for the session. Of the 240 subjects, 226 completed the session with positive cumulative earnings. Only 14 subjects (5.6 percent) completed the session with negative cumulative earnings. The mean earning of these 14 subjects was -401 points ($-\$0.53$). As the experiment turned out to be longer than anticipated, the minimum participation fee was ultimately raised to \$10. All the subjects departed the laboratory with positive earnings that exceeded \$5. Mean payoff for all the subjects was \$24.86.

Each trial was structured in the same way. Once the trial commenced, each subject was asked to independently choose and submit her departure time. Departure time decisions were made anonymously with no time pressure. In this particular environment, in order not to provide any cues for coordinating behavior, the subject's station identification number—a number between 1 and 24—was never revealed. Once all the 24 group members submitted their departure time decisions, a central computer server calculated the delays, arrival times, and travel costs which were consequently presented to the subjects. In both conditions HIGH and LOW, each subject was provided with the following information:

- The number of subjects ahead of her in each bottleneck she encountered;
- The delays and costs incurred at each phase of her trip;
- The departure time, arrival time, delays at each bottleneck, and net profit for each user.

This was accomplished by presenting a “Results” screen (see the screen in Appendix A) that consisted of a 24×5 table with rows corresponding to the 24 players (but with no identification as to the station number of the subject) ordered in terms of their departure times and colored (blue = group 1; green = group 2) according to the player's group, and 5 columns designating the

- Departure times;
- Delays at the upstream bottleneck, called Delay A;
- Delays at downstream bottleneck, called Delay B;
- Arrival times;
- Individual payoffs for the trial.

As the purpose of our study was to examine population behavior, given the game theoretic prediction of the Bottleneck Paradox, we provided the participants with complete information regarding the travel of other participants. With the development of increasingly sophisticated ATIS (Advanced Traveler Information Systems), real roadway systems are approaching this situation.

The instructions also included detailed examples of the profit calculations resulting from hypothetical departure times and bottleneck conditions (see Appendix A). The subjects played 5 practice trials for no payment to familiarize themselves with the mechanics of the game, and then played 50 additional trials with payment contingent on their performance.

B. Results

The statistical analyses presented in this section are organized as follows. First, we present means and standard deviations of the individual payoffs and use them to test the major implication of the Bottleneck Paradox. Next, we present the aggregate departure times, and for each group separately compare them to the equilibrium results. Analyses of individual behavior and learning are presented in the last subsections.

Payoffs.—Table 2 (columns 2–5) presents the means and standard deviations (in c units) of the payoffs across all 50 trials. The results are presented for group 1 (blue, $n = 8$, columns 2 and 3) and group 2 (green, $n = 16$, columns 4 and 5) and are aggregated across sessions. Within each group, they are shown separately for each of the two conditions LOW and HIGH. The right part of Table 2 (columns 6–9) displays the same statistics for the last ten trials, namely, trials 41–50. In equilibrium (see bottom row of Table 2), expansion of the upstream bottleneck results

TABLE 2—MEANS AND STANDARD DEVIATIONS OF PAYOFFS^a BY CONDITION AND GROUP

Condition	Trials 1–50				Trials 41–50			
	Group 1-blue group		Group 2-green group		Group 1-blue group		Group 2-green group	
	HIGH	LOW	HIGH	LOW	HIGH	LOW	HIGH	LOW
Mean	3.42	5.59	1.65	1.52	3.45	5.65	2.22	2.15
SD	(0.44)	(0.36)	(1.02)	(1.30)	(0.58)	(0.41)	(0.95)	(1.04)
Equilibrium	3.15	5.51	2.45	2.36	3.15	5.51	2.45	2.36

^a Experimental payoffs are expressed in units of $c = 100$ points.

TABLE 3—MEANS AND STANDARD DEVIATIONS OF PAYOFFS BY GROUP, CONDITION, AND BLOCK

Group	Condition		Block 1	Block 2	Block 3	Block 4	Block 5	Total	Theory
1 $n = 40$	HIGH	Mean	3.34	3.55	3.40	3.34	3.45	3.42	3.15
		SD	(1.55)	(1.02)	(0.81)	(0.83)	(0.56)	(1.00)	
1 $n = 40$	LOW	Mean	5.37	5.67	5.62	5.64	5.65	5.59	5.51
		SD	(1.34)	(0.36)	(0.37)	(0.39)	(0.43)	(0.67)	
2 $n = 80$	HIGH	Mean	0.47	1.82	1.78	1.97	2.22	1.65	2.45
		SD	(2.50)	(1.74)	(1.27)	(1.29)	(0.96)	(1.74)	
2 $n = 80$	LOW	Mean	0.13	1.62	1.80	1.92	2.15	1.52	2.36
		SD	(3.25)	(1.61)	(1.23)	(1.37)	(1.05)	(2.00)	

in a substantial increase in the travel cost of group 1, but only a very small decrease in the travel cost of group 2. Consequently, there is a large increase in the total system cost as capacity is increased, confirming the Bottleneck Paradox. As shown in Table 1 (right-hand column), the theoretical (mixed-strategy) equilibrium costs for group 1 are $\bar{C}_1 = 4.49$ in condition LOW and $\bar{C}_1 = 6.85$ in condition HIGH. When subtracted from the $10c$ (1,000 points) endowment for each trial, this results in equilibrium payoffs of 5.51 and 3.15 per trial for group 1 members in conditions LOW and HIGH, respectively (bottom row of Table 2). In contrast, equilibrium payoffs for group 2, whose members encounter the upstream and downstream bottlenecks, are relatively small in both conditions, 2.36 ($10.00 - 7.64$) in condition LOW and 2.45 ($10.00 - 7.55$) in condition HIGH (bottom row of Table 2).

The observed mean payoffs in Table 2 provide strong evidence in support of the equilibrium predictions. A comparison of the mean payoffs in group 1 (columns 2 and 3) shows that the subjects in condition LOW earned, on average, 63.5 percent *more* than the subjects in condition HIGH (compare 5.59 to 3.42). The corresponding change under the mixed-strategy simulation results is 74.9 percent. Using the session as the unit of analysis, the null hypothesis that the five sessions in condition HIGH and the five sessions in condition LOW are drawn from the same population was rejected by the Mann-Whitney U test ($U = 25, p < 0.01$, one-tailed test).

A comparison of the mean payoffs in group 2 (columns 4 and 5) shows that the subjects in condition LOW earned, on average, 7.9 percent *less* than the subjects in condition HIGH (1.65 versus 1.52). Using the session as the unit of analysis, the null hypothesis that the five sessions in condition HIGH and the five sessions in condition LOW are drawn from the same population could not be rejected by the Mann-Whitney U test ($p > 0.1$). This result is consistent with the model predictions that changing the capacity of the upstream bottleneck has little (discrete model) or no (flow model) impact on group 2 commuting costs.

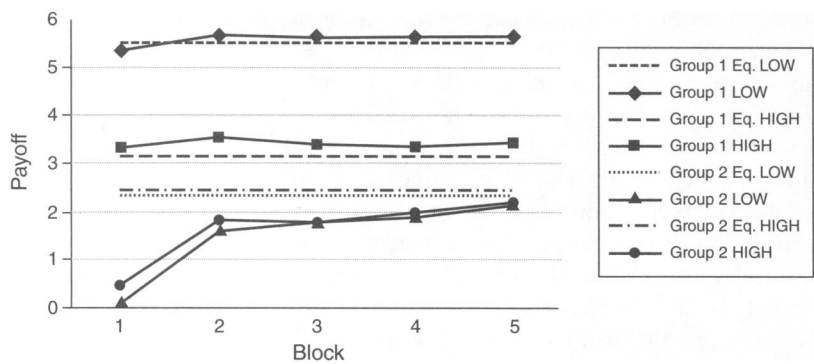


FIGURE 4A. MEAN AND EQUILIBRIUM (Eq.) PAYOFF ACROSS BLOCKS

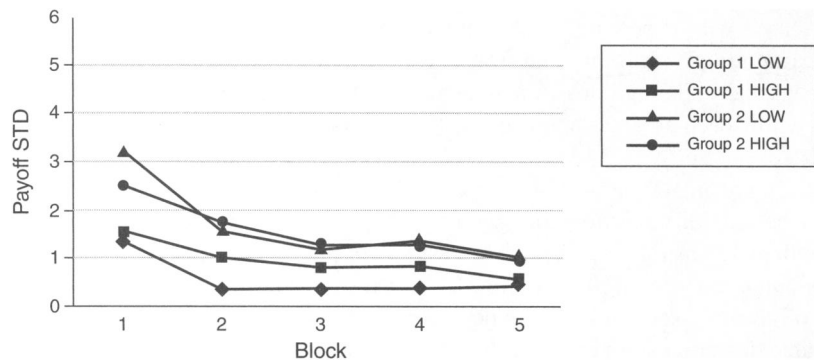


FIGURE 4B. PAYOFF STANDARD DEVIATION ACROSS BLOCKS

Table 3 and Figure 4 display the means and standard deviations of the individual payoffs by group (1 or 2), condition (LOW or HIGH), and blocks of ten trials. The results are collapsed across the five sessions. Inspection of the two top rows of Table 3 shows that, under each condition, mean payoffs for group 1 maintain approximately the same value for all five blocks: mean payoffs of the subjects who only had to pass through the downstream bottleneck are remarkably stable over the 50 trials. In contrast, the *variability* between the subjects, measured by the standard deviation, is seen to decline sharply across blocks from 1.55 in block 1 to 0.56 in block 5 of condition HIGH, and from 1.34 in block 1 to 0.43 in block 5 of condition LOW. In contrast to the stability of group 1 payoffs, mean payoffs in group 2 (rows 3 and 4) increase steadily across blocks in both conditions, with the major change occurring between blocks 1 and 2. As with group 1, but even more dramatically, the standard deviations of the individual payoffs decrease steadily across blocks from 2.50 in block 1 to 0.96 in block 5 of condition HIGH and from 3.25 in block 1 to 1.05 in block 5 of condition LOW.

Next, we tested whether payoffs change over time and whether this change depends on the group. As before, the session was used as our unit of analysis. First, for each session, we calculated the Kendall rank correlation (τ_b) between the mean group payoff and the round (1–50). A correlation of zero suggests no change in payoff over time; a positive correlation suggests an increase in payoff; and a negative correlation suggests a decrease in payoff. This calculation resulted in five correlations in condition LOW and five correlations in condition HIGH, one for

each session. A Wilcoxon test reveals that the mean correlation of group 1 was not significantly different from zero in condition LOW (mean = 0.069, $z = -0.944$, $p = 0.345$) or in condition HIGH (mean = -0.014 , $z = -0.405$, $p = 0.686$). Therefore, we conclude that there was no significant change in payoff over time. In order to test whether an interaction exists between condition and time, we conducted a Mann-Whitney test between the two sets of condition correlations. The test showed that the two did not differ significantly from each other ($U = 9$, $p = 0.465$). We then repeated the process for group 2. A (two-sided) Wilcoxon test of condition LOW showed that the mean correlation was significantly higher than zero (mean = 0.292, $z = -2.02$, $p < 0.05$). The same was true for condition HIGH (mean = 0.229, $z = -2.02$, $p < 0.05$). This significant increase in payoff is also clearly visible in Figure 4. Comparison of the two conditions using a Mann-Whitney test revealed that they were not significantly different from each other ($U = 8.5$, $p = 0.402$), suggesting that the payoff in both groups increases at the same rate.

Taken together, these results confirm our previous observation (see Table 3) that the mean payoffs of the subjects in group 1 hardly changed with experience. Moreover, they are accounted for rather well by the equilibrium solution. Mean payoffs of the subjects in group 2 (rows 3 and 4 in Table 3) steadily increased with experience in the direction of the equilibrium payoffs but stayed slightly below them. Importantly, the sharp and systematic decrease in the standard deviations of payoffs in both conditions and in both groups is in agreement with the equilibrium solution that considers all the subjects within each group as homogenous (all players within the same group earn the same).

Aggregate Departure Times.—Figure 5 exhibits the predicted and observed cumulative frequency distributions of departure time aggregated across all subjects and sessions (individual sessions are depicted as dashed lines). The results are presented separately for condition LOW (Figure 5A) and condition HIGH (Figure 5B). Because of the evidence for learning in the first two blocks of group 2 (see Table 3), the departure times are computed across trials 26–50. The only noticeable deviation from the equilibrium solutions is that the subjects in group 1 (who had to pass through a single bottleneck) departed home later than predicted; the observed cumulative frequency distributions are mostly shifted to the right of the equilibrium predicted linear line.

Comparing Figures 5A and 5B with the results displayed in Figures 3 and 4 shows that the major effect of expanding the capacity of the upstream bottleneck is, as the theory predicts, to shift the equilibrium departure time distributions for group 1 earlier: $t_1^o = 6:44$ and $t_1^e = 7:40$ in condition HIGH compared to $t_1^o = 7:18$ and $t_1^e = 7:47$ in condition LOW. The observed mean departure times of group 1 were 7:17 and 7:37 in conditions HIGH and LOW, respectively, and this 20-minute difference was shown to be significant by a Mann-Whitney test ($U = 25$, $p < 0.05$). At the same time, as predicted, the departure time distribution of group 2 is essentially unchanged.

Individual Behavior.—While the focus of this paper has been on testing the correspondence between theoretical predictions and actual behavior at the system or aggregate level, it is informative to examine the individual behavioral patterns that generate these results. Casual observation immediately casts doubt on the hypothesis that individuals, in contrast to the group in aggregate, adhere to mixed-strategy equilibrium play. Figure 6A exhibits the cumulative distribution functions across all 50 trials for each of the group 2 subjects for one representative set of data (condition HIGH, session 1). Also shown on the graph is the mixed-strategy equilibrium. While many subjects chose a distribution of departure times that bear similarity to the equilibrium distribution (for example, subjects 12 and 14), several clearly did not (for example, subjects 22 and 23). Several commuters in this session chose departure times that varied little from trial to trial.

Casting even stronger doubt on the individual use of mixed strategies is Figure 6B. This figure shows the trial-by-trial departure choices made by the four representative group 2 commuters

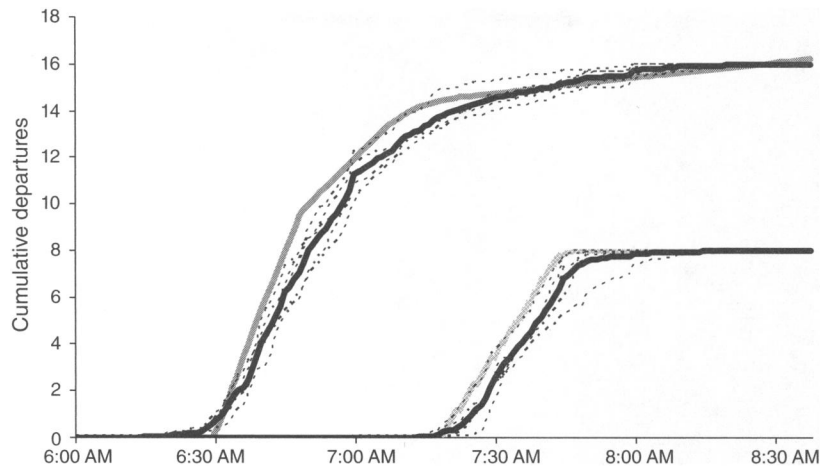


FIGURE 5A. PREDICTED AND OBSERVED DEPARTURE DISTRIBUTIONS FOR BOTH GROUPS IN CONDITION LOW

Note: Solid lines represent aggregated results, dashed are individual sessions, and gray is Nash.

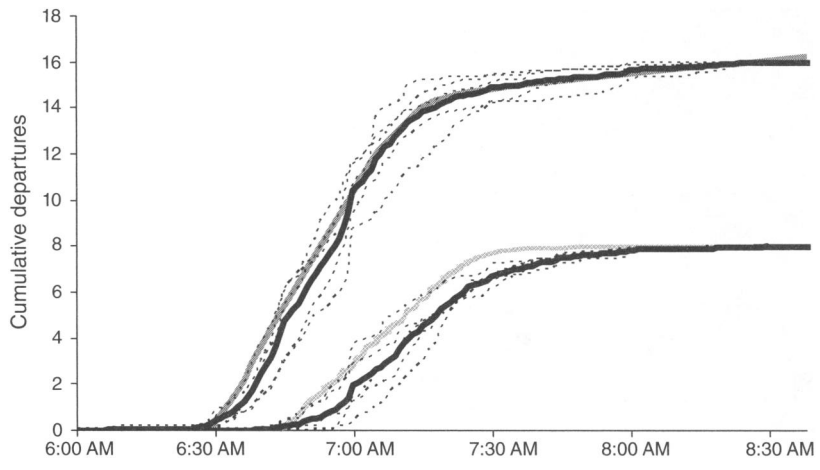


FIGURE 5B. PREDICTED AND OBSERVED DEPARTURE DISTRIBUTIONS FOR BOTH GROUPS IN CONDITION HIGH

Note: Solid lines represent aggregated results, dashed are individual sessions, and gray is Nash.

highlighted in Figure 6A. In each plot we draw two horizontal lines that correspond to the equilibrium starting time, t_2^o , and the last departure time that results in the player not being late, t_2^f (at equilibrium less than 10 percent of departures should occur after this time). Subject 12 has a distribution of departure times that appears to be close to the equilibrium in Figure 6A. However, Figure 6B shows that this subject makes only small adjustments from trial to trial, gradually exploring the support of the equilibrium distribution. On the other hand, subjects 22 and 23 do

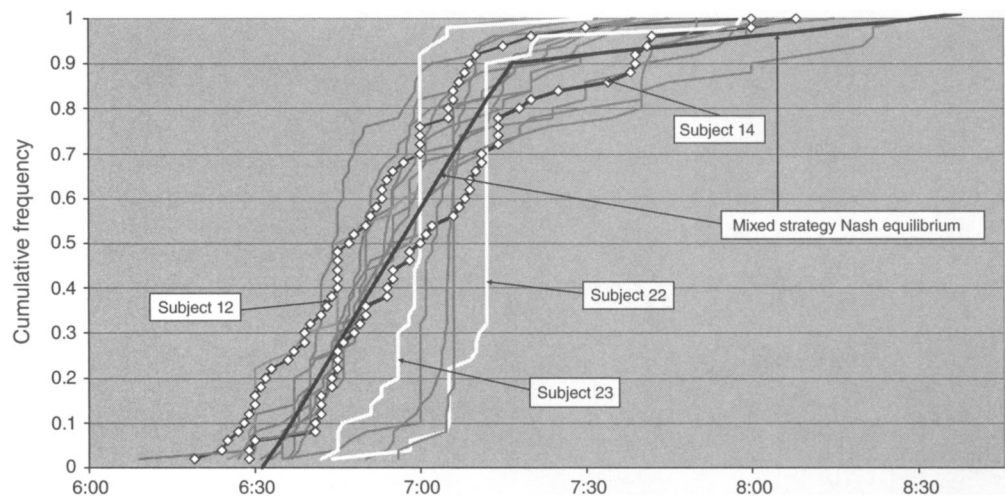


FIGURE 6A. DISTRIBUTIONS OF INDIVIDUAL DEPARTURE TIMES FOR GROUP 2 SUBJECTS IN CONDITION HIGH

Note: Shown for all 16 subjects in session 1; also shown is distribution of mixed-strategy equilibrium.

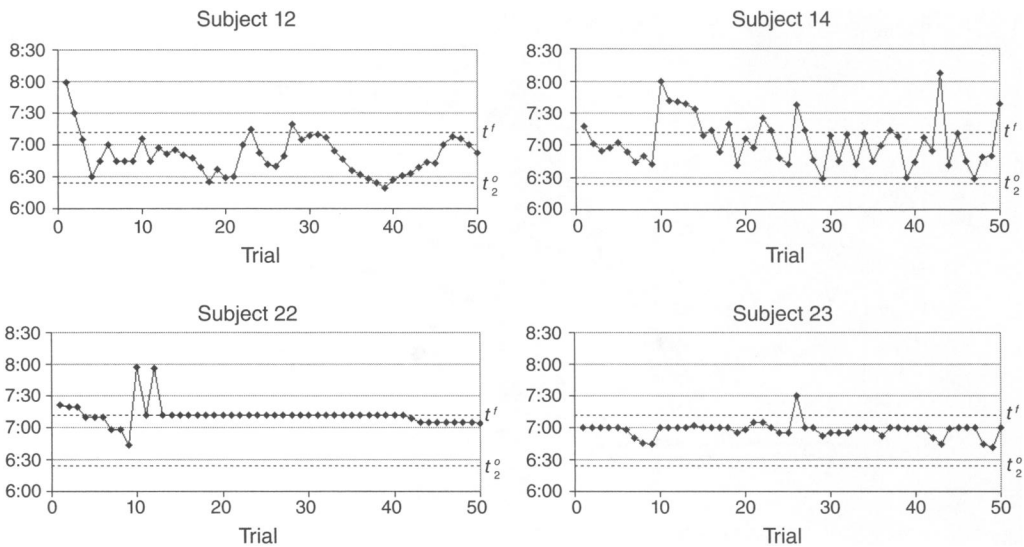


FIGURE 6B. DEPARTURE TIMES BY TRIAL FOR THE GROUP 2 SUBJECTS HIGHLIGHTED IN 6A—CONDITION HIGH (session 1)

only modest exploring and then settle on a departure time and more or less stick to it. Subject 14 is the only one of those illustrated who is a possible candidate for playing a truly mixed strategy; however, she departs after t_2^f more frequently than the model predicts.

Learning.—Because our experiment was mainly designed to test the correspondence between theory and behavior at the aggregate level, there is a limit as to the degree that this study can

discriminate between various models of individual learning behavior. However, we did compare the experimental results with the output of a simple reinforcement learning model. This model generated both individual and aggregate results that corresponded very closely to the experimental results under both the LOW and HIGH conditions.

With a couple of exceptions, our learning model is a member of the Experience-Weighted Attraction (EWA) family of models (Ho, Camerer, and Chong 2007) and is also a close derivative of the reinforcement models developed by Alvin E. Roth and Ido Erev (1998, 1995).¹¹ As in most models of this type, the payoffs and choices made in any round of a game are used to update the attractiveness (“attractions”) of the different departure times on the next round. These attractions are then transformed into probabilities that govern the choice that a player next makes.

Specifically, for commuter i , let $R_i(s_i(t), s_{-i}(t))$ denote the profit made on trial t where $s_i(t)$ and $s_{-i}(t)$ are called the *strategies* (departure times) taken by commuter i and by all others, respectively. Then, the reinforcement that i attaches to strategy j is $\pi_i^j(t) = [R_i(s_i(t), s_{-i}(t)) - R_i^o(t)]e^{-g|\Delta T|}$ where $R_i^o(t)$ is a reference point profit for commuter i on trial t and ΔT is the difference between strategy j and $s_i(t)$, the strategy actually taken by commuter i on trial t . This definition incorporates two enhancements that Erev and Roth (1998) added to their original model: the reinforcement an agent receives is measured relative to some benchmark or expectation, and that reinforcement “spills over” to strategies “near” the chosen one (“generalization”). The latter modification is particularly important as the number of strategies is very large (16,200 departure times) and most of them are never chosen in the 50-trial experiment. The parameter g controls the exponentially declining impact on departure times adjacent to the time actually chosen.¹²

For commuter i , the attraction, $A_i^j(t)$, of choosing his j th strategy after updating at trial t is $A_i^j(t) = A_i^j(t-1) + \pi_i^j(t)$. A commuter’s stochastic choice of departure time on trial $t+1$ is then governed by probabilities determined by the logistic response function, $p_i^j(t+1) = e^{\lambda A_i^j(t)} / \sum_k e^{\lambda A_i^k(t)}$, where λ is the response sensitivity (and the main parameter in the model—the higher the value of λ , the more the model tends to select the best response).

Initial attractions in model simulations were set so as to correspond with the distribution of choices made by the human subjects in the five-trial practice rounds which preceded each experiment. The performance of the learning model over the last 25 trials of a 50-trial simulation proved to be robust with respect to this initial distribution. Likewise, the model was relatively insensitive to whether the reference profits, $R_i^o(t)$, were set equal to a constant value, which, in turn, is equal to the average profits made by subjects in the practice trials of the actual experiment (and hence independent of t), or whether they were set equal to the average profit of the group in the previous trial (and hence implicitly assumed the public availability of the payoffs of all commuters as in the experiments).

After parameterizing the model,¹³ one hundred 50-trial simulations were conducted under each of the two conditions of the experiment. Figures 7A and 8A compare the cumulative distributions of departure times generated by the model under each condition over trials 26 to 50 aggregated over all 100 simulations. Also shown on these graphs are the mixed-strategy equilibrium and the distributions of first trial choices. Figures 7B and 8B show the cumulative departure time distributions

¹¹ Using the notation of Ho, Camerer, and Chong (2007), we set $N(t) = 1$, $\kappa = 1$, $\Phi = 1$ and $\delta = 0$ and used only a single parameter, λ to fit the data. Such a model is very similar to the reinforcement models of Erev and Roth (1995, 1998). Agents update their propensities to choose an action on any trial based on their own relative profit from the choice they made on the previous trial. Propensities do not depend on the profitability of choices not taken.

¹² The generalization constant was set such that departure times ten minutes from the chosen departure time received approximately 45 percent of the reinforcement of the chosen time.

¹³ Using a grid search to minimize the sum of squared deviations between the distribution of departure times of the learning model and the experimental results over trials 26 to 50, model fit was optimized with $\lambda = 1.95$ for group 2 subjects and $\lambda = 1$ for group 1 subjects in the LOW condition, and $\lambda = 2.85$ for both groups in the HIGH condition.

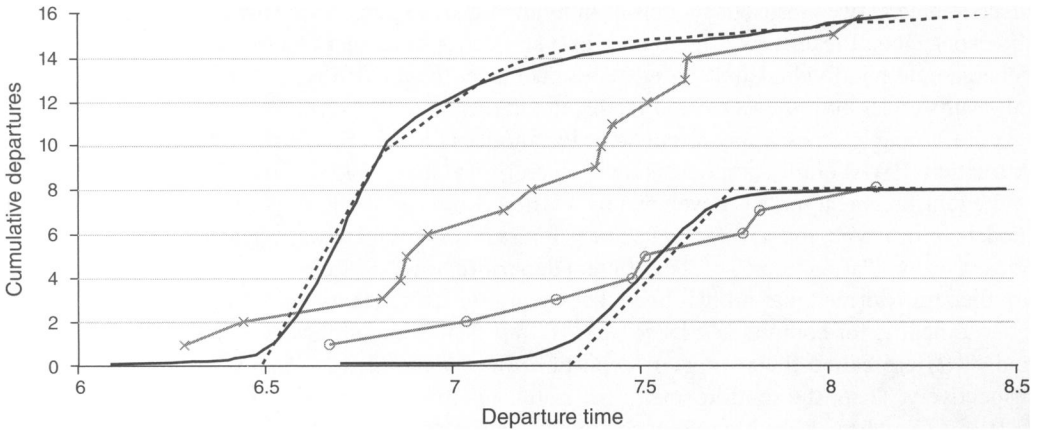


FIGURE 7A. LEARNING MODEL AVERAGE CUMULATIVE DEPARTURE TIME DISTRIBUTIONS—CONDITION LOW

Note: Dark solid lines are averages of trials 26–50 over 100 learning model simulations relative to Nash equilibrium (dotted) and trial 1 (gray) propensities.

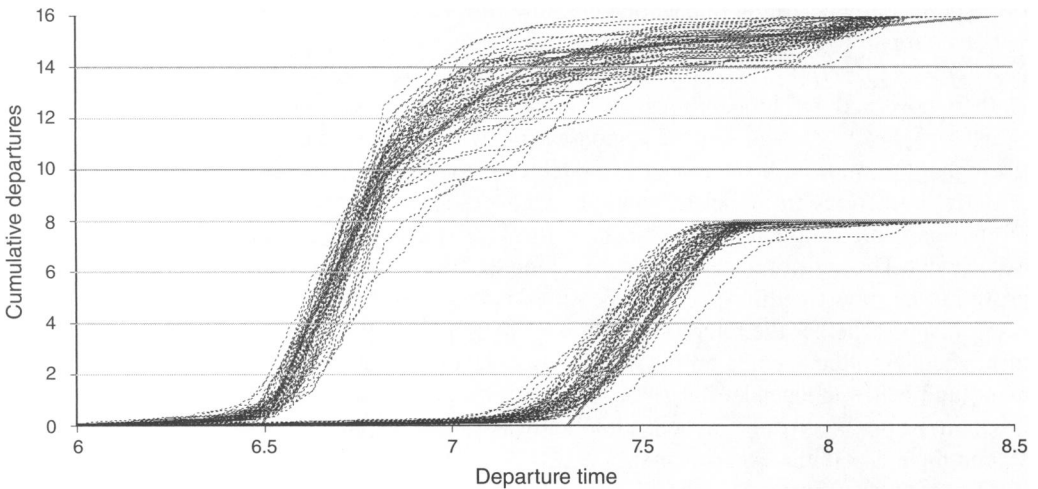


FIGURE 7B. LEARNING MODEL CUMULATIVE DEPARTURE TIME DISTRIBUTIONS BY GROUP—CONDITION LOW

Note: Shown are the departure distributions by group on trials 26–50 for each of 50 separate simulations.

for each of the first 50 of these simulations. These two graphs indicate that the learning model is very consistent. For almost all of the 100 simulations under each condition, the model moved quickly from the initial distributions to aggregate departure distributions that were strikingly consistent with both the theoretical equilibrium and the experimental distributions.

The plots of the trial-by-trial departure times of the simulated subjects (not shown) exhibit the same variety of individual strategies as shown in the experimental data in Figure 6. Many simulation agents in the learning model select a departure time and vary little from trial to trial.

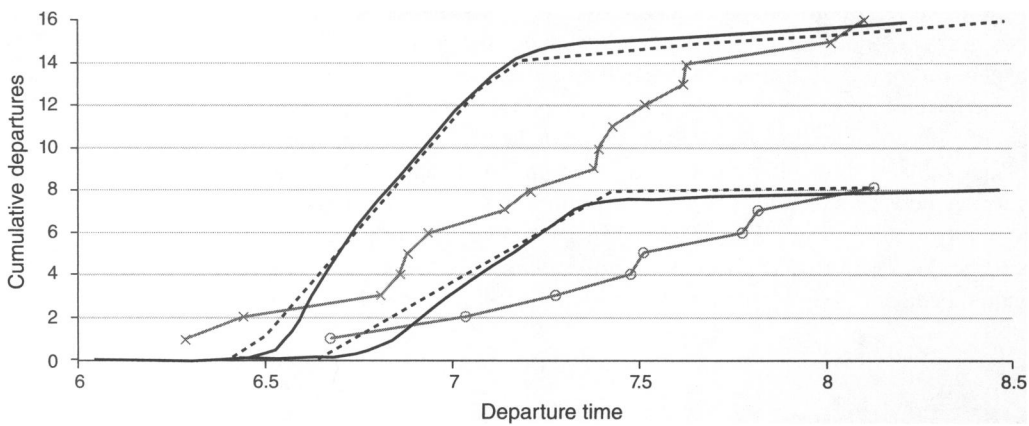


FIGURE 8A. LEARNING MODEL CUMULATIVE DEPARTURE TIME DISTRIBUTIONS—CONDITION HIGH

Note: Dark solid lines are averages of trials 26–50 over 100 learning model simulations relative to Nash (dotted) and trial 1 (gray) propensities.

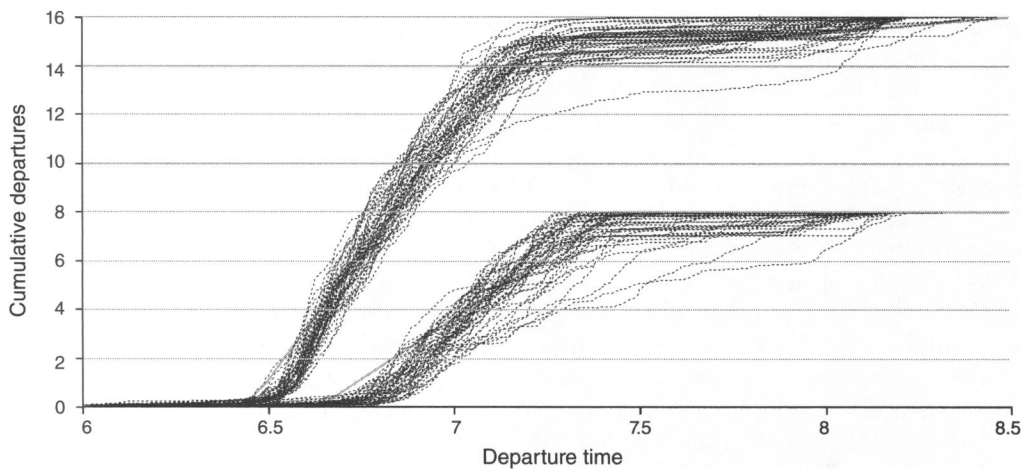


FIGURE 8B. LEARNING MODEL CUMULATIVE DEPARTURE TIME DISTRIBUTIONS BY GROUP—CONDITION HIGH

Note: Shown are the departure distributions by group on trials 26–50 for each of 50 separate simulations.

Other agents exhibit considerable fluctuation in choices, but few strategies correspond closely to the mixed-strategy equilibrium despite the remarkably close correspondence as a group.

Attempts to improve the fit of the learning model to the experimental data by allowing attractions to be dependent on the profits and choices of other agents on the previous trial¹⁴ yielded insignificant gains. As can be seen in Figures 7 and 8, the simple reinforcement model leaves

¹⁴ Such a model is closely related to “belief-based” learning models in which subjects consider the profitability of strategies not played, in addition to the profits of the strategy chosen on the previous trial.

little room for improvement in predicting the group departure times. While our search for the best fitting model was limited, it is evident that a simple reinforcement learning model in which agents respond only to their own relative payoffs captures many of the patterns of individual behavior and, most importantly for our study, the “magic”¹⁵ of group coordination in a multi-agent game. The learning model suggests, but clearly does not prove, that the remarkable ability of our subjects to quickly move, as a group, to a distribution of departure times that is in close correspondence with the theoretical equilibrium is not dependent on having full information about the outcomes of others. This observation is in line with the findings of other studies¹⁶ and will be discussed further in the concluding section of the paper. A more refined study of individual behavior is clearly an important item for future research.

IV. Discussion

Equilibrium models that invoke strong assumptions of homogeneity of group members and common knowledge of rationality acquire added value when they succeed in accounting for behavioral regularities in what seems to be counterintuitive behavior. As Arnott and Small (1994) have shown in their expository paper, several paradoxical results have been known to be present in transportation research, all of them resulting from selfish behavior of network users who ignore the negative externalities they impose on others. Experimental evidence has been reported in the last few years in an attempt to answer the question whether these transportation paradoxes are only of theoretical interest with no practical application, or are manifested in actual behavior. Morgan, Orzen, and Sefton (2009) have reported experimental evidence in support of the implications of the PKD Paradox. Even more striking is the experimental evidence that strongly supports the implications of the Braess Paradox (e.g., Rapoport et al. 2009; Morgan et al. 2009; Rapoport, Mak, and Zwick 2006). All of these studies have focused on changes in behavior resulting from exogenous changes in the cost structure imposed on the network (the PKD Paradox) or from adding or deleting one or more links in the traffic network (the Braess Paradox). Our study extends previous investigation of paradoxes in transportation networks from choice of routes when departure times are fixed to choice of departure times when the structure of the network is fixed but capacities vary. In this more challenging environment involving the coordination of departure times, behavior again supports theory and the predicted paradox emerges. As capacity of the upstream bottleneck is *increased*, social cost may *rise*.

Behavioral regularities observed in our study at the aggregate level are broadly consistent with mixed-strategy equilibrium play. With more experience in playing the iterated bottleneck game, mean payoffs approach the theoretical predictions, between-subject variability in payoff declines sharply, and departure time distributions are well approximated by the symmetric mixed-strategy equilibrium distributions. The only exception is the tendency of the subjects in group 1, whose members have to pass through a single bottleneck only, to depart from their origin slightly later than predicted. Formulating and testing explanations for this phenomenon are left for further research. Nevertheless, as predicted by the theoretical model, an attempt to improve the network by increasing the capacity of the upstream bottleneck through which only group 2 members pass results in little change in their behavior and travel costs. At the same time, this capacity expansion causes group 1 members to depart earlier and incur significantly higher

¹⁵ The term “magic” was originally used by Daniel Kahneman (1988) to describe the ability of large groups to tacitly coordinate behavior with no communication and achieve near-equilibrium outcomes.

¹⁶ For example, Ziegelmeyer et al. (2008) found an insignificant impact on choices, congestion, and costs in a transportation network with a bottleneck when subjects had information about the historical distribution of departure times of others.

costs. Overall, the investment in network capacity (increased service rate) results in network degradation. The Bottleneck Paradox is confirmed by laboratory behavior.

On the individual level, the behavioral patterns are heterogeneous, with some subjects changing their departure time quite often and others changing them hardly at all. We find no evidence that the patterns of choice made by most subjects are consistent with individual play of mixed strategy. As subjects gain experience with the bottleneck game, the variance in payoff decreases dramatically and approaches the equilibrium prediction. This low payoff variance may be viewed as a manifestation of the notion of equilibrium in which no one group member can do significantly better than the others in her group. An investigation of the dynamics of play by which subjects reach this state, using a reinforcement-based learning model, has yielded encouraging results. While this model suggests that the coordination achieved by commuters may not be a function of the information each has about the departure times of others, much work remains to be done on this important topic. Related future research should also look more closely at strategic models of commuter behavior, particularly as a function of the size of the commuter population, in an attempt to better explain the heterogeneity of individual behavior.

In conclusion, aggregated group behavior in our traffic network is accounted for very closely by the symmetric mixed-strategy equilibrium of a departure time model. This suggests that, while we cannot predict well any particular person's behavior, the Nash equilibrium accurately predicts the population choices. For many policy issues, this is what is important. Our study suggests that transportation planners should carefully consider the predictions of such models since, given the right conditions, human behavior in traffic networks can lead to capacity expansion paradoxes.

APPENDIX A

DEPARTURE TIME EXPERIMENT—INSTRUCTIONS TO PARTICIPANTS

Introduction

Welcome to an experiment on the choice of departure times. During this experiment you will be asked to make a large number of decisions and so will the other participants. Your decisions, as well as the decisions of the other participants, will determine your monetary payoff according to the rules that will be explained shortly. The money that you earn during the experiment will be paid to you in cash at the end of the session.

Please read the instructions carefully. If you have any questions, please raise your hand and someone will come to assist you.

From now on communication between the participants is prohibited. If the participants communicate with one another by any shape or form, the experiment will be terminated.

The Departure Time Task

This experiment attempts to simulate the basic features of departure time decisions that people traveling to a common destination face every day. It is fully computerized. You will be making your decisions by clicking on the appropriate buttons on the screen. A total of 24 persons are participating in this experiment (i.e., 23 participants in addition to you). During the experiment, you will participate in a series of **50** identical rounds. In each round, each participant will be asked to choose a time to depart for a common destination (for example, the time you leave for work every morning, a school class, a movie, or an airplane flight) knowing that waiting line delays may occur along the way. As in real life, there will be a value associated with reaching your destination on time and costs associated with waiting in line along the way and for either arriving too early (all participants wish to arrive by a common time; namely, 8:30 AM) or arriving too late (after 8:30 AM). These costs will be explained to you in the next section. On each trial,

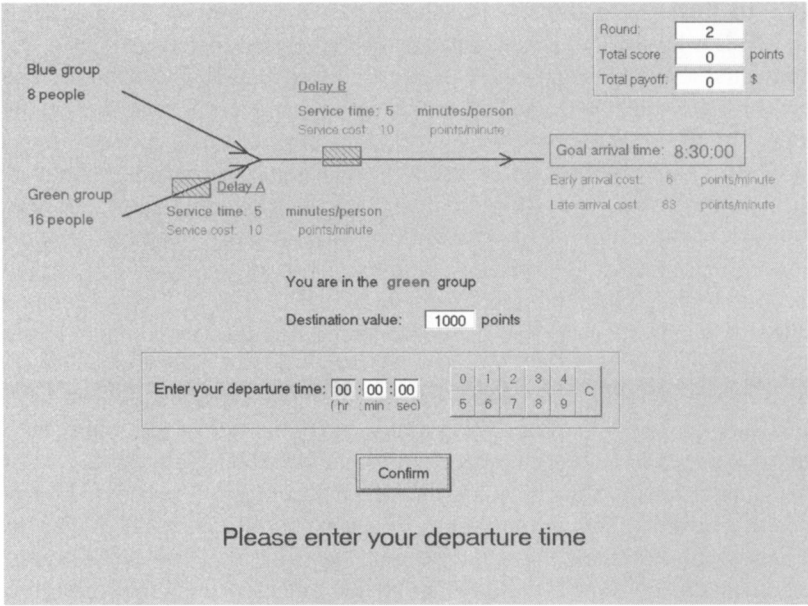


FIGURE A1. DECISION SCREEN

the participants will be paid a set amount for reaching the destination, and the costs they incur in getting there will be subtracted from their earnings. It is possible to lose money on any one trial if those costs are high.

Description of the Network

Please consult Figure A1 at the end of the instructions. Figure A1 displays a picture of the computer screen on your PC that will be presented to you on each trial. We'll now explain it. The diagram at the top of the screen illustrates the routes that the participants travel to reach their common destination in each round of the experiment. There are two groups of participants (called “Blue” and “Green”). The line on the screen just below the diagram tells you which group you belong to for all 50 rounds of the experiment. The participant illustrated in Figure A1 is a member of the “Green” group.

In traveling to their destination, members of the Blue group (8 participants) face only a single delay or bottleneck (drawn as a square on the diagram and labeled “Delay B”), whereas members of the Green group (16 participants) face an additional bottleneck (drawn as a square and labeled “Delay A”) prior to reaching bottleneck “B” that is common to both groups. (You may think of real examples of such bottlenecks such as sections of construction on roadways, ticket booths at the movie theatre, or baggage check-in and security stations at airports).

Bottlenecks slow down traffic and may cause queues to form. Participants in this experiment face a fixed service time to get through each bottleneck:

- 5 minutes at bottleneck A and
- 5 minutes at bottleneck B.

In addition, they may also have to wait in line before being served, if other network users arrived there before them and are still being served or waiting in line. For example, if two

persons arrive at bottleneck B only seconds apart, then the first person to arrive will encounter a delay of 5 minutes at B (assuming no one was there before her), and the second person will encounter a delay of nearly 10 minutes (the time waiting in line for the first person to be served plus her own service time). We assume in this experiment that it takes no time to travel between delay points or between the final delay point and the destination.

The reward and costs associated with the travel are also shown in Figure A1. They include:

- The delay costs at each bottleneck – **10** points per minute for each of bottlenecks A and B,
- The cost of arriving early – **6** points per minute;
- The cost of arriving late – **83** points per minute;
- The value associated with reaching the destination – **1,000** points.

In addition, summary information on the participant's earnings to date is shown at the top-right corner.

Procedure

Once you decide on a departure time for the current round, you will enter your chosen time using the keypad on the screen (see Figure A1 again). Note that all decisions are made with the mouse; you will not need the keyboard. To enter the hour of departure, click the number desired on the keypad on the screen. Similarly, to enter minutes and seconds, click the corresponding cells, either "min" or "sec" (the cell will turn yellow), and then click the desired numbers on the keypad. Errors can be erased by clicking the "C" button on the keypad. Clicking the "confirm" button indicates that you are satisfied with the chosen time. A confirmation text box will then appear on the screen to allow you a final opportunity to change your departure time.

Interpreting the Results

After all the participants have made their departure time choices, a results screen will appear. Figure 2 at the end of the instructions shows an example of this screen.

A summary of the delays experienced by a participant and the resulting costs and payoff are shown in the left-hand panel of the results screen (see Figure A2). The table on the right-hand panel shows the departure times and payoffs for all 24 participants. As seen in the left panel in the example shown in Figure A2, the participant departed at 7:20:00 and, as a member of the **Green** group, she had to pass through both bottlenecks. She encountered 4 persons ahead of her at delay A (the other 5 have already been served and moved on) and waited 20 minutes and 39 seconds (including her own 5-minute service time).

*This delay cost her **206.5** points (10 points per minute).*

At delay B, she encountered 5 people ahead of her and waited 30 minutes.

*This delay cost her **300** points.*

She then arrived at destination at 8:10:39 (again, note that it takes no time to travel between delay points or between the final delay point and the destination), which was about 20 minutes early.

*This early arrival cost her **116.1** points (6 points per minute).*

As a result, at the end of this round this participant was left with a payoff of **377.4** points after all costs were subtracted from the **1,000** point payoff for getting to the destination.

$$\text{Payoff} = 1,000 - 206.5 - 300 - 116.1 = 377.4 \text{ points.}$$

In the table for all participants on the right-hand side of the screen, this participant's times are highlighted in yellow as they will be during the experiment. Checking this table, you can find out the departure times of the other participants and the payoffs associated with each departure

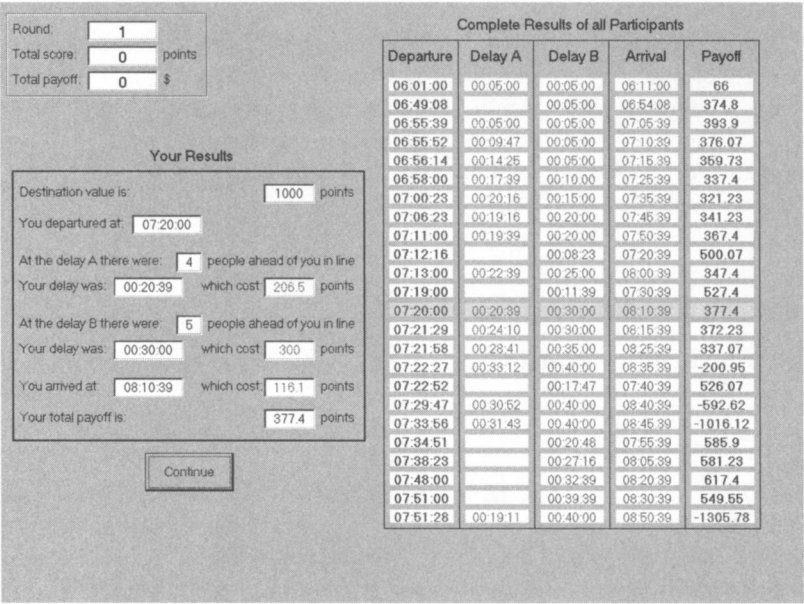


FIGURE A2. RESULTS SCREEN

time (note that the times are color coded to identify participants in the **Green** and **Blue** groups). For example, a participant in the **Green** group departed at 7:22:27 and had a delay at bottleneck A of about 33 minutes and another at B of 40 minutes and hence arrived over 5 minutes late at 8:35:39. His total payoff was a loss of over 200 points for this round largely because of the high cost of being late (83 points per minute).

You may find it helpful to think of the current experiment in terms of two groups of people choosing when to leave for an event, such as a concert, where one group already has tickets (**Blue** group) and will face only delays in getting through the security line-up at the gate to the stadium. The other group (**Green** group) has to line up to buy a ticket first, and then go through the same security gate. All the participants of both groups want to minimize time spent waiting in line and idle time spent by arriving too early. However, they also do not want to be late for the concert as this has a much higher cost for them.

End of Experiment

After completing all 50 rounds, you will see a summary screen displaying the total points you have accumulated and the corresponding earnings in dollars. (Points will be converted to money at the rate 750 points = \$1). During the experiment, the cumulative number of points and dollar payoff that you have earned will be displayed on each of the individual screens. Please remain at your desk until asked to come forward and receive payment for the experiment.

Please place the instructions on the table in front of you to indicate that you have completed reading them. The experiment will begin shortly. Initially **five (5) unpaid** practice rounds will be conducted to familiarize you with the process, and then the 50 paid rounds will follow. Please remember that no communication is allowed during the experiment. If you encounter any difficulties please raise your hand and you will be responded to by the experimenter.

REFERENCES

- Arnott, Richard, Andre de Palma, and C. Robin Lindsey. 1990. "Economics of a Bottleneck." *Journal of Urban Economics*, 27(1): 111–30.
- Arnott, Richard, Andre de Palma, and C. Robin Lindsey. 1993a. "Properties of Dynamic Traffic Equilibrium Involving Bottlenecks, Including a Paradox and Metering." *Transportation Science*, 27(2): 148–60.
- Arnott, Richard, Andre de Palma, and C. Robin Lindsey. 1993b. "A Structural Model of Peak-Period Congestion: A Traffic Bottleneck with Elastic Demand." *American Economic Review*, 83(1): 161–79.
- Arnott, Richard, Andre de Palma, and C. Robin Lindsey. 1999. "Information and Time-of-Usage Decisions in the Bottleneck Model with Stochastic Capacity and Demand." *European Economic Review*, 43(3): 525–48.
- Arnott, Richard, and Kenneth A. Small. 1994. "The Economics of Traffic Congestion." *American Scientist*, 82: 446–55.
- Avineri, Erel. 2006. "Measuring and Simulating Altruistic Behaviour in Group Travel Choice Decisions." Unpublished.
- Avineri, Erel, and Joseph N. Prashker. 2005. "Sensitivity to Travel Time Variability: Travelers' Learning Perspective." *Transportation Research Part C: Emerging Technologies*, 13(2): 157–83.
- Braess, Dietrich. 1968. "Über ein Paradoxon der Verkehrsplanung." *Unternehmensforschung*, 12: 258–68.
- Daganzo, Carlos F. 1985. "The Uniqueness of a Time-dependent Equilibrium Distribution of Arrivals at a Single Bottleneck." *Transportation Science*, 19(1): 29–37.
- Denant-Boëmont, Laurent, and Romain Petiot. 2003. "Information Value and Sequential Decision-Making in a Transport Setting: An Experimental Study." *Transportation Research Part B: Methodological*, 37(4): 365–86.
- Erev, Ido, and Alvin E. Roth. 1998. "Predicting How People Play Games: Reinforcement Learning in Experimental Games with Unique, Mixed Strategy Equilibria." *American Economic Review*, 88(4): 848–81.
- Gabuthy, Yannick, Matthie Neveu, and Laurent Denant-Boëmont. 2006. "The Coordination Problem in a Structural Model of Peak-Period Congestion: An Experimental Study." *Review of Network Economics*, 5(2): 273–98.
- Helbing, Dirk. 2004. "Dynamic Decision Behavior and Optimal Guidance through Information Services: Models and Experiments." In *Human Behaviour and Traffic Networks*, ed. Michael Schreckenberg and Reinhard Selten, 47–95. New York: Springer.
- Helbing, Dirk, Martin Schonhof, and Daniel Kern. 2002. "Volatile Decision Dynamics: Experiments, Stochastic Description, Intermittency Control and Traffic Optimization." *New Journal of Physics*, 4(1): 33.1–33.16.
- Hendrickson, Chris, and George Kocur. 1981. "Schedule Delay and Departure Time Decisions in a Deterministic Model." *Transportation Science*, 15(1): 62–77.
- Ho, Teck H., Colin F. Camerer, and Juin-Kuan Chong. 2007. "Self-Tuning Experience Weighted Attraction Learning in Games." *Journal of Economic Theory*, 133(1): 177–98.
- Iida, Yasunori, Takamasa Akiyama, and Takashi Uchida. 1992. "Experimental Analysis of Dynamic Route Choice Behavior." *Transportation Research Part B: Methodological*, 26(1): 17–32.
- Kahneman, Daniel. 1988. "Experimental Economics: A Psychological Perspective." In *Bounded Rational Behavior in Experimental Games and Markets*, ed. Reinhard Tietz, W. Albers, and Reinhard Selten, 11–18. New York: Springer.
- Knight, Frank H. 1924. "Some Fallacies in the Interpretation of Social Cost." *Quarterly Journal of Economics*, 38(4): 582–606.
- Knodel, Walter. 1969. *Graphentheoretische Methoden und ihre Anwendungen*. Berlin: Springer.
- Kuwahara, Masao. 1990. "Equilibrium Queuing Patterns at a Two-Tandem Bottleneck during the Morning Peak." *Transportation Science*, 24(3): 217–29.
- Mahmassani, Hani S., and Gang-Len Chang. 1987. "On Boundedly Rational User Equilibrium in Transportation Systems." *Transportation Science*, 21(2): 89–99.
- Mahmassani, Hani S., Gang-Len Chang, and Robert Herman. 1986. "Individual Decisions and Collective Effects in a Simulated Traffic System." *Transportation Science*, 20(4): 258–71.
- Morgan, John, Henrik Orzen, and Martin Sefton. 2009. "Network Architecture and Traffic Flows: Experiments on the Pigou-Knight-Downs and Braess Paradoxes." *Games and Economic Behavior*, 66(1): 348–72.
- Murchland, John D. 1970. "Braess's Paradox of Traffic Flow." *Transportation Research*, 4(4): 391–94.

- Otsubo, Hironori, and Amnon Rapoport. 2008. "Vickrey's Model of Traffic Congestion Discretized." *Transportation Research Part B: Methodological*, 42(10): 873–89.
- Rapoport, Amnon, Vincent Mak, and Rami Zwick. 2006. "Navigating Congested Networks with Variable Demand: Experimental Evidence." *Journal of Economic Psychology*, 27(5): 648–66.
- Rapoport, Amnon, Tamar Kugler, Subhasish Dugar, and Eyrán J. Gisches. 2008. "Braess Paradox in The Laboratory: An Experimental Study of Route Choice in Traffic Networks with Asymmetric Costs." In *Decision Modeling and Behavior in Uncertain and Complex Environments*, ed. Tamar Kugler, J. Cole Smith, Terry Connolly, and Young J. Son, 309–37. New York: Springer.
- Rapoport, Amnon, Tamar Kugler, Subhasish Dugar, and Eyrán J. Gisches. 2009. "Choice of Routes in Congested Traffic Networks: Experimental Tests of the Braess Paradox." *Games and Economic Behavior*, 65(2): 538–71.
- Roth, Alvin E., and Ido Erev. 1995. "Learning in Extensive-Form Games: Experimental Data and Simple Dynamic Models in the Intermediate Term." *Games and Economic Behavior*, 8(1): 164–212.
- Schneider, Kerstin, and Joachim Weimann. 2004. "Against All Odds: Nash Equilibria in a Road Pricing Experiment." In *Human Behaviour and Traffic Networks*, ed. Michael Schreckenberg and Reinhard Selten, 133–53. New York: Springer.
- Selten, Reinhard, Thorsten Chmura, Thomas Pitz, and Michael Schreckenberg. 2007. "Commuters Route Choice Behaviour." *Games and Economic Behavior*, 58(2): 394–406.
- Selten, Reinhard, Michael Schreckenberg, Thorsten Chmura, Thomas Pitz, Sebastian Kube, Sigurour F. Hafstein, Ronald Chrobok, Andreas Pottmeier, and Joachim Wahle. 2004. "Experimental Investigation of Day-to-Day Route-Choice Behaviour and Network Simulations of Autobahn Traffic in North Rhine-Westphalia." In *Human Behaviour and Traffic Networks*, ed. Michael Schreckenberg and Reinhard Selten, 1–21. New York: Springer.
- Smith, Mike J. 1983. "The Existence and Calculation of Traffic Equilibria." *Transportation Research Part B: Methodological*, 17(4): 291–303.
- Vickrey, William S. 1969. "Congestion Theory and Transport Investment." *American Economic Review*, 59(2): 251–60.
- Ziegelmeyer, Anthony, Frédéric Koessler, Kene Boun My, Laurent Denant-Boèmont. 2008. "Road Traffic Congestion and Public Information: An Experimental Investigation." *Journal of Transport Economics and Policy*, 42(1): 43–82.